

or service specifically designed and used to find security defects. There are various ways to discover potential vulnerabilities. You can either go for manual testing or automated testing. Manual testing is done by white hat hackers, whereas an automated pen test is performed by software.

The target for the pen test can be a white box, a black box, or a gray box. The background and system information about a white box target is provided to the tester beforehand. A black box is a target about which only basic information is provided. Sometimes no information is provided, apart from the company name for which the tester must do the test. A gray box target is a combination of the two, and only limited information is provided to the tester.

Benefits of penetration testing

A pen test helps in an in-depth analysis of the security infrastructure and has many benefits when it comes to protecting a system from data losses. Here are some of the significant benefits.

Reveals vulnerabilities: A pen test reveals the weakness in the system or the network infrastructure. A report obtained after the test indicates the areas where the system is lacking. It also suggests measures that can be taken to improve both the software and hardware of a computer system. These recommendations help improve the overall security of the computer system and the network.

Ensures ongoing trust: A data breach can negatively affect a company's reputation and lead to losses in the future, as customers may lose faith in it. It is vital to conduct regular pen tests to ensure security and maintain the confidence of customers in your brand. Systematic security reviews reassure customers, suppliers, partners and stakeholders.

Analysis of infrastructure and potential risks: The job of a pen tester is to exploit vulnerabilities using

methods that a hacker would use. This includes accessing sensitive information like passwords, contact information, bank details, etc. The tester can also execute system commands and gain remote access to the system. Only a tester can tell you about all the known risks and how likely these are to be exploited if not fixed at the earliest.

Use cases

Most big organisations use pen testing for their large and complex business critical operations. They also use it for custom components specific to their organisation. Pen testing becomes necessary for software that handles sensitive data. Particular sectors like the government, as well as the healthcare and financial services require strong security measures considering the nature of the information they store.

Some of the use cases for penetration testing are:

- Tracking data transmitted across a wire
- Monitoring data stored in a file
- Checking for secret passwords stored in a file
- Checking if the error page exposes any data
- Checking URLs for sensitive data
- Checking for DOS attack strategies like repeating the same action, changing expected data types, etc
- Checking for an XML injection attack
- Checking for HTML script injection attacks
- Checking for buffer overflow
- Checking for spoofing attacks like changing the MAC address and the IP address

Types of pen testing

Penetration tests intentionally compromise an organisation's security system to find potential weaknesses. If there is enough protection, the security team alerts the same during the trial. But if the system fails the test, it is considered exposed to risk. Such computer systems are vulnerable to

attacks and can disclose confidential information. There are different ways to perform pen tests.

Open-box pen test: The hacker or tester is provided with some basic information ahead of time. This information can be regarding the target company and its previous security measures.

Closed-box pen test: This is also known as a single-blind test. In this type of pen testing, hackers are given no information whatsoever. The only thing they know is the name of the company.

Covert pen test: This type of test is also called a double-blind pen test. In this type of testing, no one in the organisation is aware that a pen test is going on. This includes all the IT and the security professionals expected to respond to the attack. For this type of test, the hacker must have details like the scope and target of the test in writing beforehand so as not to land into any trouble later on.

External pen test: In this type of test, the hacker tries to hack using the company's external-facing technology like its website or network servers. In this case, hackers generally conduct the attack remotely as they may not be allowed to enter the company's premises.

Internal pen test: In this type of pen test, the hacker performs the test from within the company's internal network. This means the hacker is allowed to use the company's resources to carry out the attack. This is done to see how much damage could happen behind the company's firewall if certain employees don't do their due diligence.

There is also a range of penetration testing to help uncover vulnerabilities across the entire IT infrastructure. Some of these pen tests are listed below.

Web app test: This is used to find any potential holes in the security of Web application software.

Network test: This exposes the weakness or vulnerability within the host network or the network devices connected to that host.